Exam Formula Sheet

Epi 202: Probability

$$\begin{aligned}\operatorname{Var}\Big(\underset{\sim}{a}\cdot\underset{\sim}{X}\Big) &= \operatorname{Var}\left(\sum_{i=1}^{n} a_i X_i\right)\\ &= \underset{\sim}{a}^{\top}\operatorname{Var}\Big(\underset{\sim}{X}\Big)\underset{\sim}{a}\\ &= \sum_{i=1}^{n}\sum_{j=1}^{n} a_i a_j \operatorname{Cov}(X_i, X_j)\end{aligned}$$

$$\mathrm{E}[Y] = \mathrm{E}[\mathrm{E}[Y \mid X]]$$

$$\mathrm{E}[Y \mid Z] = \mathrm{E}[\mathrm{E}[Y \mid X, Z] \mid Z]$$

$$\operatorname{Var}(Y) = \mathrm{E}[\operatorname{Var}(Y \mid X)] + \operatorname{Var}(\mathrm{E}[Y \mid X])$$

$$\operatorname{Cov}(Y, Z) = \mathrm{E}[\operatorname{Cov}(Y, Z \mid X)] + \operatorname{Cov}(\mathrm{E}[Y \mid X], \mathrm{E}[Z \mid X])$$

Epi 203: Statistical inference

$$\mathcal{L}(\theta) \overset{\text{def}}{=} \mathrm{p}(\underset{\sim}{X} = \underset{\sim}{x} | \Theta = \theta)$$

$$\ell \overset{\text{def}}{=} \log\Big\{\mathcal{L}(\underset{\sim}{x}|\theta)\Big\}$$

$$\ell' \overset{\text{def}}{=} \frac{\partial}{\partial\theta}\ell(\underset{\sim}{x}|\theta)$$

$$\ell'' \overset{\text{def}}{=} \frac{\partial}{\partial\underset{\sim}{\theta}}\frac{\partial}{\partial\underset{\sim}{\theta}^{\top}}\ell(\underset{\sim}{x}|\underset{\sim}{\theta})$$

$$\ell''_{ij} = \frac{\partial}{\partial\theta_i}\frac{\partial}{\partial\theta_j}\ell(\underset{\sim}{X} = \underset{\sim}{x}|\theta)$$

$$I \overset{\text{def}}{=} -\ell''(\underset{\sim}{x}|\underset{\sim}{\theta})$$

$$\mathcal{I} \overset{\text{def}}{=} \mathrm{E}\Big[I(\underset{\sim}{x}|\theta)\Big]$$

$$\hat{\theta}_{ML} \mathrel{\dot\sim} \mathrm{N}\left(\theta, \Big[\mathcal{I}(\underset{\sim}{\theta})\Big]^{-1}\right)$$

For one parameter θ_k:

$$\mathrm{CI}_{1-\alpha}(\theta_k) = \Big[\hat{\theta}_k \pm z_{1-\frac{\alpha}{2}}\,\widehat{\mathrm{SE}}\big(\hat{\theta}_k\big)\Big]$$

$$Z_k \overset{\text{def}}{=} \frac{\hat{\theta}_k - \theta_{k,0}}{\widehat{\mathrm{SE}}\big(\hat{\theta}_k\big)} \mathrel{\dot\sim} \mathrm{N}(0,1) \qquad \text{under } H_0 : \theta_k = \theta_{k,0} \qquad z_k = \text{observed } Z_k$$

$$\begin{aligned}p\text{-value} &= 2\Pr(|Z| \geq |z_k|), \quad Z \sim \mathrm{N}(0,1)\\ &= 2(1 - \Phi(|z_k|))\end{aligned}$$

Sta 108: Linear regression

$$t_k \stackrel{\text{def}}{=} \frac{\hat{\beta}_k - \beta_{k,0}}{\widehat{\text{SE}}\left(\hat{\beta}_k\right)} \mathrel{\dot\sim} t_{n-p} \qquad \text{under } H_0 : \beta_k = \beta_{k,0} \qquad t_k^{\text{obs}} = \text{observed } t_k$$

$$\text{CI}_{1-\alpha}(\beta_k) = \left[\hat{\beta}_k \pm t_{n-p}\left(1 - \frac{\alpha}{2}\right) \widehat{\text{SE}}\left(\hat{\beta}_k\right)\right]$$

Let $T_{n-p} \sim t_{n-p}$.

$$\begin{aligned} p\text{-value} &= 2\Pr\left(|T_{n-p}| \geq |t_k^{\text{obs}}|\right) \\ &= 2\left(1 - F_{t_{n-p}}(|t_k^{\text{obs}}|)\right) \end{aligned}$$

$$\text{CI}_{1-\alpha}\left(\mu\left(\underset{\sim}{x}^*\right)\right) = \left[\hat{\mu}\left(\underset{\sim}{x}^*\right) \pm t_{n-p}\left(1 - \frac{\alpha}{2}\right) \widehat{\text{SE}}\left(\hat{\mu}\left(\underset{\sim}{x}^*\right)\right)\right]$$

Y^* denotes a new observation (not in the training data), with corresponding covariate pattern $\underset{\sim}{x}^*$. Let $\hat{Y}^* \stackrel{\text{def}}{=} \hat{\mu}\left(\underset{\sim}{x}^*\right)$.

$$\text{PI}_{1-\alpha}\left(Y^* | \underset{\sim}{x}^*\right) = \left[\hat{Y}^* \pm t_{n-p}\left(1 - \frac{\alpha}{2}\right) \widehat{\text{SE}}\left(Y^* - \hat{Y}^*\right)\right]$$

$$\widehat{\text{SE}}\left(Y^* - \hat{Y}^*\right) = \hat{\sigma}\sqrt{1 + \left(\underset{\sim}{x}^*\right)^\top (\mathbf{X}'\mathbf{X})^{-1} \underset{\sim}{x}^*}$$

$$\text{Var}\left(Y^* - \hat{Y}^*\right) = \sigma^2\left(1 + \left(\underset{\sim}{x}^*\right)^\top (\mathbf{X}'\mathbf{X})^{-1} \underset{\sim}{x}^*\right)$$

Let $\hat{\Sigma} \stackrel{\text{def}}{=} \widehat{\text{Var}}\left(\hat{\beta}\right) = \hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}$.

$$\widehat{\text{Var}}\left(\hat{\mu}\left(\underset{\sim}{x}\right)\right) = \underset{\sim}{x}^\top \hat{\Sigma} \underset{\sim}{x}$$

Let $\Delta\mu\left(\underset{\sim}{x}, \underset{\sim}{x}^*\right) = \mu\left(\underset{\sim}{x}\right) - \mu\left(\underset{\sim}{x}^*\right)$, and let $\Delta\underset{\sim}{x} = \underset{\sim}{x} - \underset{\sim}{x}^*$; then:

$$\widehat{\text{Var}}\left(\widehat{\Delta\mu}\left(\underset{\sim}{x}, \underset{\sim}{x}^*\right)\right) = \Delta\underset{\sim}{x}^\top \hat{\Sigma} \Delta\underset{\sim}{x}$$

Epi 204: Generalized linear models

Generalized linear models have three components:

1. The **outcome distribution** family: $\text{p}(Y|\mu(\underset{\sim}{x}))$
2. The **link function**: $g(\mu(\underset{\sim}{x})) = \eta(\underset{\sim}{x})$
3. The **linear component**: $\eta(\underset{\sim}{x}) = \underset{\sim}{x} \cdot \beta$

$$_\omega(\underset{\sim}{x}, \underset{\sim}{x}^*) = \exp\left\{(\Delta\underset{\sim}{x}) \cdot \underset{\sim}{\beta}\right\}$$

Estimates of odds ratios from 2x2 contingency tables

$$\hat{\theta} = \frac{ad}{bc}$$

Summary of logistic regression definitions and results

Odds and log-odds

These association-measure results are covered in Measures of Association for Binary Outcomes[1].

Odds (def[2])

$$\omega \stackrel{\text{def}}{=} \frac{\Pr(A)}{\Pr(\neg A)}$$

Conditional odds (def[3])

$$\omega(A|B) \stackrel{\text{def}}{=} \frac{\Pr(A|B)}{\Pr(\neg A|B)}$$

[1] binary-outcome-associations.qmd
[2] binary-outcome-associations.qmd#def-odds
[3] binary-outcome-associations.qmd#def-c-odds

Odds function (def[4])

$$\text{odds}\{\pi\} \stackrel{\text{def}}{=} \frac{\pi}{1-\pi}$$

Probability to odds (thm[5], cor[6])

$$\omega = \frac{\pi}{1-\pi} = \text{odds}\{\pi\}$$

Simplified odds expressions; odds of a non-event (thm[7], cor[8])

$$\text{odds}\{\pi\} = \frac{1}{\pi^{-1}-1} = (\pi^{-1}-1)^{-1}, \qquad \omega(\neg A) = \frac{1-\pi}{\pi} = \pi^{-1}-1$$

Odds ratio (def[9])

$$(\omega_1, \omega_2) \stackrel{\text{def}}{=} \frac{\omega_1}{\omega_2}$$

OR as ratio of probability ratios (thm[10])

$$(\omega_1, \omega_2) = \frac{\omega_1}{\omega_2} = \frac{\left(\frac{\pi_1}{1-\pi_1}\right)}{\left(\frac{\pi_2}{1-\pi_2}\right)}$$

Odds ratios are reversible, marginally and conditionally (thm[11], thm[12])

$$_{\omega}(A|B) = \ _{\omega}(B|A), \qquad _{\omega}(A|B,C) = \ _{\omega}(B|A,C)$$

Inverse-odds and probability recovery
Odds to probability (thm[13])

$$\pi = \frac{\omega}{1+\omega}$$

Inverse-odds function (def[14]; recovers probability by cor[15])

$$\text{invodds}\{\omega\} \stackrel{\text{def}}{=} \frac{\omega}{1+\omega}, \qquad \pi = \text{invodds}\{\omega\}$$

Simplified inverse-odds (lem[16])

$$\text{invodds}\{\omega\} = \frac{1}{1+\omega^{-1}} = (1+\omega^{-1})^{-1}$$

One minus inverse-odds, and its complement (lem[17], cor[18])

$$1-\pi = \frac{1}{1+\omega}, \qquad 1+\omega = \frac{1}{1-\pi}$$

Log-odds (logit) and expit
Log-odds, and log-odds from probability (def[19], thm[20])

$$\eta \stackrel{\text{def}}{=} \log\{\omega\}, \qquad \eta = \log\left\{\frac{\pi}{1-\pi}\right\}$$

[4] binary-outcome-associations.qmd#def-odds-fn
[5] binary-outcome-associations.qmd#thm-prob-to-odds
[6] binary-outcome-associations.qmd#cor-oddsf-to-odds
[7] binary-outcome-associations.qmd#thm-odds-simplified
[8] binary-outcome-associations.qmd#cor-odds-neg
[9] binary-outcome-associations.qmd#def-OR
[10] binary-outcome-associations.qmd#thm-or-ratio-ratio
[11] binary-outcome-associations.qmd#thm-or-swap
[12] binary-outcome-associations.qmd#thm-conditional-OR-swap
[13] binary-outcome-associations.qmd#thm-odds-to-prob
[14] binary-outcome-associations.qmd#def-inv-odds
[15] binary-outcome-associations.qmd#cor-invodds-pi
[16] binary-outcome-associations.qmd#lem-invodds-simplified
[17] binary-outcome-associations.qmd#lem-one-minus-odds-inv
[18] binary-outcome-associations.qmd#cor-inverse-odds-nonevent
[19] binary-outcome-associations.qmd#def-logodds
[20] binary-outcome-associations.qmd#thm-logodds-pi

Logit function, and expanded form (def[21], thm[22])

$$\text{logit}(\pi) \overset{\text{def}}{=} \log\{\text{odds}\{\pi\}\} = \log\left\{\frac{\pi}{1-\pi}\right\}$$

Log-odds equals logit (cor[23])

$$\eta = \text{logit}\{\pi\}$$

Odds and probability from log-odds (lem[24], thm[25])

$$\omega = \exp\{\eta\}, \qquad \pi = \frac{\exp\{\eta\}}{1+\exp\{\eta\}}$$

Expit / inverse-logit function, and equivalent expressions (def[26], thm[27])

$$\text{expit}(\eta) \overset{\text{def}}{=} \text{invodds}\{\exp\{\eta\}\} = \frac{\exp\{\eta\}}{1+\exp\{\eta\}} = (1+\exp\{-\eta\})^{-1}$$

Probability as expit (thm[28])

$$\pi = \text{expit}\{\eta\} \tag{1}$$

Logit and expit are inverses (thm[29])

$$\text{logit}\{\text{expit}\{\eta\}\} = \eta \qquad \text{expit}\{\text{logit}\{\pi\}\} = \pi$$

$$\overbrace{\left[\pi \overset{\text{def}}{=} \Pr\left(Y=1|\underset{\sim}{X}=\underset{\sim}{x}\right)\right] \underset{\frac{\omega}{1+\omega}}{\overset{\frac{\pi}{1-\pi}}{\rightleftarrows}} \left[\omega \overset{\text{def}}{=} \text{odds}\left\{Y=1|\underset{\sim}{X}=\underset{\sim}{x}\right\}\right] \underset{\exp\{\eta\}}{\overset{\log\{\omega\}}{\rightleftarrows}} \left[\eta(\underset{\sim}{x}) \overset{\text{def}}{=} \eta\left(Y=1|\underset{\sim}{X}=\underset{\sim}{x}\right)\right]}^{\text{logit}(\pi)}$$
$$\underbrace{\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad}_{\text{expit}(\eta)}$$

Figure 1: Diagram of logistic regression[30] link and inverse link functions[31]

Rare events

Odds minus probability (thm[32])

$$\omega - \pi = \frac{\pi^2}{1-\pi}, \quad \text{where } \omega = \frac{\pi}{1-\pi}$$

OR as RR times a correction factor (thm[33])

$$(\omega_1, \omega_2) = \rho(\pi_1, \pi_2) \cdot \frac{1-\pi_2}{1-\pi_1}$$

When π_1 and π_2 are both small, the correction factor is close to 1, so the odds ratio[34] is close to the risk ratio[35].

[21] binary-outcome-associations.qmd#def-logit-fn
[22] binary-outcome-associations.qmd#thm-logit-function
[23] binary-outcome-associations.qmd#cor-logodds-logit
[24] binary-outcome-associations.qmd#lem-odds-from-logodds
[25] binary-outcome-associations.qmd#thm-prob-from-logodds
[26] binary-outcome-associations.qmd#def-expit
[27] binary-outcome-associations.qmd#thm-expit-expressions
[28] binary-outcome-associations.qmd#thm-expit-prob-logodds
[29] binary-outcome-associations.qmd#thm-logit-expit
[32] binary-outcome-associations.qmd#thm-odds-minus-probs
[33] binary-outcome-associations.qmd#thm-rare-disease-or-rr
[34] binary-outcome-associations.qmd#def-OR
[35] binary-outcome-associations.qmd#def-RR

Derivatives

	π	ω	η	$\underset{\sim}{x}$	$\underset{\sim}{\beta}$
π	1	$(1+\omega)^2$	$\dfrac{(1+\omega)^2}{\omega}$	undef	undef
ω	$(1-\pi)^2$	1	$\dfrac{1}{\omega}$	undef	undef
η	$\pi(1-\pi)$	ω	1	undef	undef
$\underset{\sim}{x}$	$\underbrace{\underset{\sim}{\beta}\pi(1-\pi)}_{p\times 1}$	$\underbrace{\underset{\sim}{\beta}\omega}_{p\times 1}$	$\underbrace{\underset{\sim}{\beta}}_{p\times 1}$	$\underbrace{\mathbb{I}}_{p\times p}$	$\mathbf{0}_{p\times p}$
$\underset{\sim}{\beta}$	$\underbrace{x\pi(1-\pi)}_{p\times 1}$	$\underbrace{x\omega}_{p\times 1}$	$\underbrace{x}_{p\times 1}$	$\mathbf{0}_{p\times p}$	$\underbrace{\mathbb{I}}_{p\times p}$

Column labels indicate the numerators of the derivatives[36]; row labels indicate the denominators (dimensions and details in tbl[37]).

Derivative of expit (cor[38])

$$\text{expit}'\{\eta\} = \text{expit}\{\eta\}(1-\text{expit}\{\eta\})$$

Logistic regression model

Logistic regression (def[39])

$$Y_i|\underset{\sim}{X}_i \sim_{\perp\!\!\!\perp} \text{Ber}(\pi(\underset{\sim}{X}_i)), \qquad \text{logit}\Big\{\pi(\underset{\sim}{x})\Big\} = \underset{\sim}{x}^\top\underset{\sim}{\beta}$$

Throughout this summary, $\underset{\sim}{\beta} = (\beta_0, \beta_1, \ldots, \beta_p)$ is the $(p+1)\times 1$ coefficient vector (including the intercept), $\mathbf{X}$ is the $n\times(p+1)$ design matrix whose i^{th} row is $\underset{\sim i}{x}^\top$, and $\underset{\sim}{\varepsilon}$ is the $n\times 1$ error vector with components $\varepsilon_i \overset{\text{def}}{=} y_i - \pi_i$.

Log-likelihood and score function

Log-likelihood component (lem[40])

$$\ell_i(\pi_i) = y_i\eta_i - \log\{1+\omega_i\}$$

Score function as sum of components (lem[41], thm[42])

$$\underset{\sim}{\ell}'(\underset{\sim}{\beta}) = \sum_{i=1}^n \underset{\sim}{\ell}_i'(\underset{\sim}{\beta}), \qquad \underset{\sim}{\ell}_i'(\underset{\sim}{\beta}) = \underset{\sim i}{x}\,\varepsilon_i$$

Score function[43] (thm[44])

$$\underset{\sim}{\ell}'(\underset{\sim}{\beta}) = \sum_{i=1}^n \underset{\sim i}{x}\,\varepsilon_i = \underbrace{\underset{\sim}{\mathbf{X}}^\top}_{(p+1)\times n}\underbrace{\underset{\sim}{\varepsilon}}_{n\times 1}$$

Maximum likelihood estimation

Estimating equation (eq[45]) The MLE[46] $\underset{\sim}{\hat\beta}$ solves the score equation[47]:

$$\underbrace{\underset{\sim}{\ell}'(\underset{\sim}{\hat\beta})}_{(p+1)\times 1} = \sum_{i=1}^n \underbrace{\underset{\sim i}{x}}_{(p+1)\times 1}\underbrace{\Big(y_i - \text{expit}\Big\{\underset{\sim i}{x}^\top\underset{\sim}{\hat\beta}\Big\}\Big)}_{1\times 1} = \mathbf{0}_{(p+1)\times 1}$$

This estimating equation has no closed-form solution in general; $\underset{\sim}{\hat\beta}$ must be computed by numerical iteration.

[36] https://morrison-lab.github.io/mds/calculus-derivatives.html#def-derivative
[37] logistic-regression.qmd#tbl-logistic-deriv-matrix
[38] logistic-regression.qmd#cor-deriv-expit
[39] logistic-regression.qmd#def-logistic-regression
[40] logistic-regression.qmd#lem-logistic-loglik-component
[41] logistic-regression.qmd#lem-score-logistic
[42] logistic-regression.qmd#thm-logistic-score-comp
[43] https://morrison-lab.github.io/sds/intro-MLEs.html#def-score
[44] logistic-regression.qmd#thm-logistic-score-fn
[45] logistic-regression.qmd#eq-score-logistic
[46] https://morrison-lab.github.io/sds/intro-MLEs.html#def-mle
[47] https://morrison-lab.github.io/sds/intro-MLEs.html#def-score-equation

Hessian (eq[48])

$$\underbrace{\ell''(\beta)}_{(p+1)\times(p+1)} = -\underbrace{\mathbf{X}^\top}_{(p+1)\times n}\underbrace{\mathbf{D}}_{n\times n}\underbrace{\mathbf{X}}_{n\times(p+1)}, \qquad \mathbf{D} \overset{\text{def}}{=} \operatorname{diag}(\operatorname{Var}\left(Y_i | X_i = x_i\right)) = \operatorname{diag}(\pi_i(1-\pi_i))$$

Concavity (rem[49]): the Hessian[50] is negative semi-definite for every β (negative definite when $\mathbf{X}$ has full column rank), so the log-likelihood[51] is concave[52] and every solution of the score equation is a global maximum.

Newton-Raphson update (Newton-Raphson method[53])

$$\underbrace{\hat{\beta}^*}_{(p+1)\times 1} \leftarrow \underbrace{\hat{\beta}^*}_{(p+1)\times 1} - \underbrace{\left(\ell''\left(y; \hat{\beta}^*\right)\right)^{-1}}_{(p+1)\times(p+1)} \underbrace{\ell'\left(y; \hat{\beta}^*\right)}_{(p+1)\times 1}$$

One-sample MLE for odds (thm[54])

$$\hat{\omega} = \frac{x}{n-x}$$

Wald tests and confidence intervals for coefficients

Wald test statistic for $H_0 : \beta_k = \beta_{k,0}$ (CLT for MLEs[55])

$$z_k = \frac{\hat{\beta}_k - \beta_{k,0}}{\widehat{\operatorname{SE}}\left(\hat{\beta}_k\right)}, \qquad z_k \dot{\sim} N(0,1) \text{ under } H_0$$

95% confidence intervals for β_k and e^{β_k} (MLE[56] invariance)

$$\hat{\beta}_k \pm 1.96 \cdot \widehat{\operatorname{SE}}\left(\hat{\beta}_k\right), \qquad e^{\beta_k} \in \left(e^{\hat{\beta}_k - 1.96\cdot\widehat{\operatorname{SE}}(\hat{\beta}_k)},\ e^{\hat{\beta}_k + 1.96\cdot\widehat{\operatorname{SE}}(\hat{\beta}_k)}\right)$$

For $k \neq 0$, e^{β_k} is an odds ratio[57]; for $k = 0$, e^{β_0} is the baseline odds[58].

Odds ratios in logistic regression

General OR formula (lem[59])

$$\omega(x, x^*) = \exp\left\{\eta(x) - \eta(x^*)\right\}$$

Difference in log-odds; OR in terms of $\Delta\eta$ (def[60], cor[61])

$$\Delta\eta \overset{\text{def}}{=} \eta(x) - \eta(x^*), \qquad \omega(x, x^*) = \exp\{\Delta\eta\}$$

$\Delta\eta$ from covariates (lem[62])

$$\Delta\eta = (x - x^*) \cdot \beta$$

Difference in covariate patterns; $\Delta\eta$ from Δx (def[63], cor[64])

$$\Delta x \overset{\text{def}}{=} x - x^*, \qquad \Delta\eta = \Delta x \cdot \beta$$

OR in terms of Δx (thm[65])

$$\omega(x, x^*) = \exp\left\{(\Delta x) \cdot \beta\right\}$$

[48] logistic-regression.qmd#eq-logistic-hess
[49] logistic-regression.qmd#rem-hessian-nsd
[50] https://morrison-lab.github.io/mds/vector-calculus.html#def-hessian
[51] https://morrison-lab.github.io/sds/intro-MLEs.html#def-loglik
[52] https://morrison-lab.github.io/mds/optimization.html#def-strictly-convex
[53] https://morrison-lab.github.io/sds/intro-MLEs.html#sec-newton-raphson
[54] binary-outcome-associations.qmd#thm-est-odds
[55] https://morrison-lab.github.io/sds/intro-MLEs.html#thm-dist-mle
[56] https://morrison-lab.github.io/sds/intro-MLEs.html#def-mle
[57] binary-outcome-associations.qmd#def-OR
[58] binary-outcome-associations.qmd#def-odds
[59] logistic-regression.qmd#lem-logistic-OR
[60] logistic-regression.qmd#def-difflogodds
[61] logistic-regression.qmd#cor-logistic-OR
[62] logistic-regression.qmd#lem-diff-logodds
[63] logistic-regression.qmd#def-diff-covs
[64] logistic-regression.qmd#cor-diff-logodds
[65] logistic-regression.qmd#thm-logistic-OR

Log OR equals $\Delta\eta$ (cor[66])

$$\log\Big\{ {}_{\omega}(\underset{\sim}{x}, \underset{\sim}{x}^*)\Big\} = \Delta\eta$$

Inference for log-odds and odds ratios

Estimated SE of log-odds (thm[67])

$$\widehat{\mathrm{Var}}\Big(\hat{\eta}(\underset{\sim}{x})\Big) = \underset{1\times(p+1)}{\underset{\sim}{x}^\top} \underset{(p+1)\times(p+1)}{\hat{\underset{\sim}{\Sigma}}} \underset{(p+1)\times 1}{\underset{\sim}{x}}, \qquad \widehat{\mathrm{SE}}\Big(\hat{\eta}(\underset{\sim}{x})\Big) = \sqrt{\underset{\sim}{x}^\top \hat{\underset{\sim}{\Sigma}} \underset{\sim}{x}}$$

Estimated SE of $\Delta\hat{\eta}$ (thm[68])

$$\widehat{\mathrm{Var}}(\Delta\hat{\eta}) = \underset{1\times(p+1)}{\underbrace{\Delta\underset{\sim}{x}^\top}} \underset{(p+1)\times(p+1)}{\hat{\underset{\sim}{\Sigma}}} \underset{(p+1)\times 1}{\underbrace{(\Delta\underset{\sim}{x})}}, \qquad \widehat{\mathrm{SE}}(\Delta\hat{\eta}) = \sqrt{\Delta\underset{\sim}{x}^\top \hat{\underset{\sim}{\Sigma}} (\Delta\underset{\sim}{x})}$$

CI for an odds ratio (eq[69]; details in sec[70]): exponentiate the endpoints of the CI for $\Delta\eta$:

$${}_{\omega}(\underset{\sim}{x}, \underset{\sim}{x}^*) \in (e^L, e^R), \qquad (L, R) = \widehat{\Delta\eta} \mp 1.96 \cdot \widehat{\mathrm{SE}}\Big(\widehat{\Delta\eta}\Big)$$

Odds ratios in study designs

Disease and exposure odds ratios (def[71], def[72])

$${}_{\omega}(D|E) \overset{\text{def}}{=} \frac{\omega(D|E)}{\omega(D|\neg E)}, \qquad {}_{\omega}(E|D) \overset{\text{def}}{=} \frac{\omega(E|D)}{\omega(E|\neg D)}$$

Case-control studies (section[73]): outcome-stratified sampling breaks estimation of risks[74], risk ratios[75], and the components of the disease odds ratio (def[76]), but the exposure odds ratio (def[77]) is estimable, and it equals the disease odds ratio by reversibility (thm[78]).

2×2 **table shortcut** (table[79]; cell counts a, b = exposed events and non-events, c, d = unexposed events and non-events):

$$\hat{\omega}(Event|Exposed) = \frac{a}{b}, \qquad \hat{\omega}(Event|\neg Exposed) = \frac{c}{d}, \qquad {}_{\omega}(Exposed, \neg Exposed) = \frac{ad}{bc}$$

Model fit and comparison

Details in sec[80]; here q is the number of distinct covariate patterns, p is the number of β parameters, and ℓ_{full} is the saturated-model log-likelihood.

Deviance test (def[81])

$$D \overset{\text{def}}{=} 2(\ell_{\text{full}} - \ell(\hat{\beta}; \mathbf{x})), \qquad D \mathrel{\dot\sim} \chi^2(q-p)$$

Hosmer-Lemeshow test (def[82]; g groups by predicted probability)

$$X^2 \overset{\text{def}}{=} \sum_{c=1}^{g} \frac{(o_c - e_c)^2}{e_c} \mathrel{\dot\sim} \chi^2(g-2)$$

Pearson chi-squared GOF test (sum of squared Pearson residuals, def[83])

$$X^2 = \sum_{k=1}^{q} X_k^2 \mathrel{\dot\sim} \chi^2(q-p)$$

[66] logistic-regression.qmd#cor-log-or
[67] logistic-regression.qmd#thm-est-se-logodds
[68] logistic-regression.qmd#thm-est-se-diff-logodds
[69] logistic-regression.qmd#eq-ci-OR-exp
[70] logistic-regression.qmd#sec-or-inference
[71] binary-outcome-associations.qmd#def-disease-OR
[72] binary-outcome-associations.qmd#def-exposure-OR
[73] binary-outcome-associations.qmd#sec-CC-studies
[74] binary-outcome-associations.qmd#def-risk-epi
[75] binary-outcome-associations.qmd#def-RR
[76] binary-outcome-associations.qmd#def-disease-OR
[77] binary-outcome-associations.qmd#def-exposure-OR
[78] binary-outcome-associations.qmd#thm-or-swap
[79] binary-outcome-associations.qmd#tbl-2x2-generic
[80] logistic-regression.qmd#sec-gof
[81] logistic-regression.qmd#def-deviance-test
[82] logistic-regression.qmd#def-hosmer-lemeshow
[83] logistic-regression.qmd#def-pearson-residual

AIC and BIC [lower is better] (AIC[84] (Akaike 1974); BIC[85] (Schwarz 1978))

$$\text{AIC} = -2\ell(\hat{\theta}) + 2p, \qquad \text{BIC} = -2\ell(\hat{\theta}) + p\log\{n\}$$

Likelihood ratio test (def[86]; for nested models with $p_0 < p_1$ parameters — here ℓ_1 and ℓ_0 are the two fitted models' log-likelihoods[87], not the saturated model[88]'s)

$$\Lambda \overset{\text{def}}{=} 2(\ell_1(\hat{\theta}_1) - \ell_0(\hat{\theta}_0)), \qquad \Lambda \mathrel{\dot{\sim}} \chi^2(p_1 - p_0) \text{ under } H_0$$

by Wilks' theorem[89].

Residual-based diagnostics

Residuals are computed per covariate pattern k (grouped data); h_k denotes the leverage of pattern k (def[90]).

Response residual (def[91])

$$e_k \overset{\text{def}}{=} \bar{y}_k - \hat{\pi}(x_k)$$

Pearson residual (and standardized version) (def[92], def[93])

$$X_k \overset{\text{def}}{=} \frac{\bar{y}_k - \hat{\pi}_k}{\sqrt{\hat{\pi}_k(1-\hat{\pi}_k)/n_k}}, \qquad r_{P_k} \overset{\text{def}}{=} \frac{X_k}{\sqrt{1-h_k}}$$

Deviance residual (and standardized version) (def[94])

$$d_k \overset{\text{def}}{=} \text{sign}(y_k - n_k\hat{\pi}_k)\left\{\sqrt{2[\ell_{\text{full}}(x_k) - \ell(\hat{\beta}; x_k)]}\right\}, \qquad r_{D_k} \overset{\text{def}}{=} \frac{d_k}{\sqrt{1-h_k}}$$

Cook's distance (def[95])

$$D_k \overset{\text{def}}{=} \frac{(r_{P_k})^2}{p} \cdot \frac{h_k}{1-h_k}$$

Leverage (def[96]): h_k is the k-th diagonal element of the weighted hat matrix[97] (with $\mathbf{X}$ here the $q \times p$ design matrix[98] of covariate patterns):

$$\underset{q\times q}{\mathbf{H}} \overset{\text{def}}{=} \underbrace{\mathbf{W}^{1/2}}_{q\times q}\ \underset{q\times p}{\mathbf{X}}\left(\underbrace{\mathbf{X}^\top\mathbf{W}\mathbf{X}}_{p\times p}\right)^{-1}\underbrace{\mathbf{X}^\top}_{p\times q}\ \underbrace{\mathbf{W}^{1/2}}_{q\times q}, \qquad \underset{q\times q}{\mathbf{W}} \overset{\text{def}}{=} \text{diag}(n_k\,\hat{\pi}_k\,(1-\hat{\pi}_k))$$

Comparing probabilities

Risk difference (def[99])

$$\delta(\pi_1, \pi_2) \overset{\text{def}}{=} \pi_1 - \pi_2$$

Risk ratio (def[100])

$$\rho(\pi_1, \pi_2) \overset{\text{def}}{=} \frac{\pi_1}{\pi_2}$$

Relative risk difference (def[101]; equals RR minus 1 by thm[102])

$$\xi(\pi_1, \pi_2) \overset{\text{def}}{=} \frac{\delta(\pi_1, \pi_2)}{\pi_2} = \rho(\pi_1, \pi_2) - 1$$

[84] Linear-models-overview.qmd#def-aic
[85] Linear-models-overview.qmd#def-bic
[86] logistic-regression.qmd#def-lrt
[87] https://morrison-lab.github.io/sds/intro-MLEs.html#def-loglik
[88] Linear-models-overview.qmd#def-saturated-model
[89] https://morrison-lab.github.io/sds/intro-MLEs.html#thm-wilks
[90] logistic-regression.qmd#def-leverage-glm
[91] logistic-regression.qmd#def-response-residual-logistic
[92] logistic-regression.qmd#def-pearson-residual
[93] logistic-regression.qmd#def-standardized-pearson-residual
[94] logistic-regression.qmd#def-deviance-residual-logistic
[95] logistic-regression.qmd#def-cooks-distance-glm
[96] logistic-regression.qmd#def-leverage-glm
[97] Linear-models-overview.qmd#def-hat-matrix
[98] https://morrison-lab.github.io/mds/linear-algebra-projections.html#def-design-matrix
[99] binary-outcome-associations.qmd#def-RD
[100] binary-outcome-associations.qmd#def-RR
[101] binary-outcome-associations.qmd#def-RRD
[102] binary-outcome-associations.qmd#thm-rrd-vs-rr

Marginal risks from logistic regression

Conditional predicted risk (def[103])

$$\pi(a, z) \stackrel{\text{def}}{=} \Pr(Y = 1 \mid A = a, Z = z)$$

Conditional RD and RR (def[104], def[105])

$$\mathrm{RD}(z) \stackrel{\text{def}}{=} \pi(1, z) - \pi(0, z), \qquad \mathrm{RR}(z) \stackrel{\text{def}}{=} \frac{\pi(1, z)}{\pi(0, z)}$$

Observed marginal risk (def[106]; standardization form by thm[107])

$$\pi(a) \stackrel{\text{def}}{=} \Pr(Y = 1 \mid A = a) = \mathrm{E}[\pi(a, Z) \mid A = a]$$

Potential mean and causal marginal risk (def[108], def[109]; identification form by thm[110])

$$\mu_a \stackrel{\text{def}}{=} \mathrm{E}[Y^a], \qquad \pi_a \stackrel{\text{def}}{=} \Pr(Y^a = 1) = \mathrm{E}[\pi(a, Z)]$$

No confounding: observed = causal (cor[111]; the two estimands[112] are contrasted in rem[113])

$$Z \perp\!\!\!\perp A \implies \pi(a) = \pi_a$$

G-computation estimator (def[114]; consistent for π_a by thm[115])

$$\hat{\pi}_a \stackrel{\text{def}}{=} \frac{1}{n} \sum_{i=1}^{n} \hat{\pi}_{a,i}, \qquad \hat{\pi}_{a,i} \stackrel{\text{def}}{=} \hat{\pi}(a, z_i)$$

Predictive margins (def[116]; full-sample form def[117] is consistent for π_a by cor[118])

$$\widehat{\mathrm{PM}}(a \mid S) \stackrel{\text{def}}{=} \frac{1}{|S|} \sum_{i \in S} \hat{\pi}_{a,i}, \qquad \widehat{\mathrm{PM}}(a) \stackrel{\text{def}}{=} \frac{1}{n} \sum_{i=1}^{n} \hat{\pi}_{a,i} = \hat{\pi}_a$$

Exposed-subgroup predictive margin (def[119]; consistent for $\pi(a)$ by thm[120]; equals $\bar{Y}_a$ exactly by lem[121])

$$\hat{\pi}_a(a) \stackrel{\text{def}}{=} \frac{1}{n_a} \sum_{i : A_i = a} \hat{\pi}_{a,i} = \bar{Y}_a$$

Observed marginal RD and RR (def[122], def[123])

$$\mathrm{OMRD} \stackrel{\text{def}}{=} \pi(1) - \pi(0), \qquad \mathrm{OMRR} \stackrel{\text{def}}{=} \frac{\pi(1)}{\pi(0)}$$

Causal marginal RD and RR (def[124], def[125])

$$\mathrm{CMRD} \stackrel{\text{def}}{=} \pi_1 - \pi_0, \qquad \mathrm{CMRR} \stackrel{\text{def}}{=} \frac{\pi_1}{\pi_0}$$

[103] logistic-regression.qmd#def-cond-predicted-risk
[104] logistic-regression.qmd#def-cond-rd
[105] logistic-regression.qmd#def-cond-rrr
[106] logistic-regression.qmd#def-observed-marginal-risk
[107] logistic-regression.qmd#thm-observed-marginal-risk
[108] logistic-regression.qmd#def-potential-mean
[109] logistic-regression.qmd#def-causal-marginal-risk
[110] logistic-regression.qmd#thm-causal-marginal-risk
[111] logistic-regression.qmd#cor-observed-equals-causal
[112] https://morrison-lab.github.io/sds/estimation.html#def-estimand
[113] logistic-regression.qmd#rem-observed-vs-causal-marginal-risk
[114] logistic-regression.qmd#def-g-computation
[115] logistic-regression.qmd#thm-g-computation-consistency
[116] logistic-regression.qmd#def-predictive-margins
[117] logistic-regression.qmd#def-full-sample-predictive-margin
[118] logistic-regression.qmd#cor-full-sample-pm
[119] logistic-regression.qmd#def-subgroup-predictive-margin
[120] logistic-regression.qmd#thm-subgroup-pm
[121] logistic-regression.qmd#lem-subgroup-pm-proportion
[122] logistic-regression.qmd#def-observed-marginal-rd
[123] logistic-regression.qmd#def-observed-marginal-rr
[124] logistic-regression.qmd#def-causal-marginal-rd
[125] logistic-regression.qmd#def-causal-marginal-rr

Bootstrap SE and CI for marginal contrasts (def[126]; see Bootstrap Confidence Intervals[127] and exm[128]): resample the data with replacement B times; refit the logistic model and recompute the marginal contrast on each resample; the bootstrap SE is the standard deviation[129] of the B estimates, and a 95% percentile CI uses their 2.5th and 97.5th percentiles.

Collapsibility (def[130]): an estimand θ is collapsible with respect to Z if

$$\theta \stackrel{\text{def}}{=} \mathrm{E}[\theta(Z)]$$

Risk differences[131] are collapsible; risk ratios[132] and odds ratios[133] are not collapsible in general (thm[134]; counterexample in exm[135]).

Survival analysis

Probability distribution functions

Table 1: Probability distribution functions

Name	Symbols	Definition
Probability density function (PDF)[136]	$\mathrm{f}(t), \mathrm{p}(t)$	$\mathrm{p}(T = t)$
Cumulative distribution function (CDF)[137]	$\mathrm{F}(t), \mathrm{P}(t)$	$\mathrm{P}(T \leq t)$
Survival function[138]	$\mathrm{S}(t), \bar{\mathrm{F}}(t)$	$\mathrm{P}(T > t)$
Hazard function[139]	$\lambda(t), \mathrm{h}(t)$	$\mathrm{p}(T = t \mid T \geq t)$
Cumulative hazard function[140]	$\Lambda(t), \mathrm{H}(t)$	$\int_{u=-\infty}^{t} \lambda(u)du$
Log-hazard function[141]	$\eta(t)$	$\log\{\lambda(t)\}$

Diagram of survival distribution function relationships

$$\mathrm{f}(t) \xleftarrow[\mathrm{S}(t)\lambda(t)]{-S'(t)} \mathrm{S}(t) \xleftarrow{\exp\{-\Lambda(t)\}} \Lambda(t) \xleftarrow{\int_{u=0}^{t} \lambda(u)du} \lambda(t) \xleftarrow{\exp\{\eta(t)\}} \eta(t)$$

$$\mathrm{f}(t) \xrightarrow[\int_{u=t}^{\infty} \mathrm{f}(u)du]{\mathrm{f}(t)/\lambda(t)} \mathrm{S}(t) \xrightarrow[-\log \mathrm{S}(t)]{} \Lambda(t) \xrightarrow[\Lambda'(t)]{} \lambda(t) \xrightarrow[\log\{\lambda(t)\}]{} \eta(t)$$

Survival likelihood contributions, assuming non-informative censoring

$$\begin{aligned} \mathrm{p}(Y = y, D = d) &= [\mathrm{f}_T(y)]^d [\mathrm{S}_T(y)]^{1-d} \\ &= [\lambda_T(y)]^d [\mathrm{S}_T(y)] \end{aligned}$$

Nonparametric time-to-event distribution estimators

$$\hat{\lambda}_i = \frac{d_i}{r_i}$$

$$\hat{\kappa}_i = 1 - \hat{\lambda}_i = \frac{r_i - d_i}{r_i}$$

$$\hat{\mathrm{S}}_{\mathrm{KM}}(t) \stackrel{\text{def}}{=} \prod_{\{i:\ t_i \leq t\}} \hat{\kappa}_i$$

$$\hat{\Lambda}_{NA}(t) \stackrel{\text{def}}{=} \sum_{\{i:\ t_i \leq t\}} \hat{\lambda}_i$$

Proportional hazards model structure

[126] logistic-regression.qmd#def-bootstrap-marginal-rd
[127] https://morrison-lab.github.io/sds/bootstrap.html#sec-bootstrap-ci
[128] logistic-regression.qmd#exm-wcgs-marginal-rd
[129] https://morrison-lab.github.io/pds/variance-covariance.html#def-sd
[130] logistic-regression.qmd#def-collapsibility
[131] binary-outcome-associations.qmd#def-RD
[132] binary-outcome-associations.qmd#def-RR
[133] binary-outcome-associations.qmd#def-OR
[134] logistic-regression.qmd#thm-collapsibility
[135] logistic-regression.qmd#exm-collapsibility
[136] https://morrison-lab.github.io/pds/random-variables.html#def-pdf
[137] https://morrison-lab.github.io/pds/random-variables.html#def-cdf
[138] intro-to-survival-analysis.qmd#def-surv-fn
[139] intro-to-survival-analysis.qmd#def-hazard
[140] intro-to-survival-analysis.qmd#def-cuhaz
[141] proportional-hazards-models.qmd#def-loghaz

Formula	Description
$\lambda(t \mid \underset{\sim}{x}) = \lambda_0(t) \cdot \theta_\lambda(\underset{\sim}{x})$	Proportional hazards assumption[142]
$\Lambda(t \mid \underset{\sim}{x}) = \Lambda_0(t) \cdot \theta_\lambda(\underset{\sim}{x})$	Cumulative hazard factorization[143]
$\eta(t \mid \underset{\sim}{x}) = \eta_0(t) + \Delta\eta(\underset{\sim}{x})$	Log-hazard decomposition[144]
$\Delta\eta(\underset{\sim}{x}) = \underset{\sim}{x} \cdot \underset{\sim}{\beta} = \beta_1 x_1 + \cdots + \beta_p x_p$	Linear predictor[145]
$\theta_\lambda(\underset{\sim}{x}) = \exp\left\{\Delta\eta(\underset{\sim}{x})\right\}$	Hazard multiplier[146]
$\lambda(t \mid \underset{\sim}{x} : \underset{\sim}{x}^*) \overset{\text{def}}{=} \frac{\lambda(t \mid \underset{\sim}{x})}{\lambda(t \mid \underset{\sim}{x}^*)}$	Hazard ratio definition[147]
$\lambda(t \mid \underset{\sim}{x} : \underset{\sim}{x}^*) = \exp\left\{\Delta\eta(\underset{\sim}{x}) - \Delta\eta(\underset{\sim}{x}^*)\right\} = \exp\left\{(\underset{\sim}{x} - \underset{\sim}{x}^*) \cdot \underset{\sim}{\beta}\right\}$	Hazard ratio formula[148]

Proportional hazards model partial likelihood formula:

Definition 0.1 (Cox PH partial likelihood)**.**

$$\mathcal{L}_i^*(\underset{\sim}{\beta}) \overset{\text{def}}{=} \frac{\theta_\lambda(\underset{\sim(i)}{x})}{\sum_{k \in R(t_i)} \theta_\lambda(\underset{\sim k}{x})}$$

$$\mathcal{L}^*(\underset{\sim}{\beta}) \overset{\text{def}}{=} \prod_{\{i:\ d_i = 1\}} \mathcal{L}_i^*(\underset{\sim}{\beta})$$

where, for the distinct event times $t_1 < \cdots < t_K$ of the data set:

- (i) is the subject who fails at t_i (unique under the no-ties assumption);
- $R(t_i) \overset{\text{def}}{=} \{j : Y_j \geq t_i\}$ is the risk set[a] at t_i (where $Y_j = \min(T_j, C_j)$ is subject j's observed follow-up time) — the subjects still under observation just before t_i;
- d_i is the number of events at t_i (so $d_i = 1$ when there are no ties);
- $\theta_\lambda(\underset{\sim}{x}) \overset{\text{def}}{=} \exp\left\{\underset{\sim}{x}^\top \underset{\sim}{\beta}\right\}$ is the hazard ratio (risk score[b]) for covariate pattern $\underset{\sim}{x}$.

[a] intro-to-survival-analysis.qmd#def-risk-set
[b] proportional-hazards-models.qmd#def-risk-score

Proportional hazards model baseline cumulative hazard estimator:

Definition 0.2 (Breslow estimator of the baseline cumulative hazard)**.**

$$\hat{\Lambda}_0(t) \overset{\text{def}}{=} \sum_{t_i \leq t} \frac{d_i}{\sum_{k \in R(t_i)} \theta_\lambda(\underset{\sim k}{x})}$$

where, for the distinct event times $t_1 < \cdots < t_K$ of the data set:

- d_i is the number of events at t_i;
- $R(t_i) \overset{\text{def}}{=} \{j : Y_j \geq t_i\}$ is the risk set[a] at t_i (where $Y_j = \min(T_j, C_j)$ is subject j's observed follow-up time) — the subjects still under observation just before t_i;
- $\theta_\lambda(\underset{\sim}{x}) \overset{\text{def}}{=} \exp\left\{\underset{\sim}{x}^\top \underset{\sim}{\beta}\right\}$ is the hazard ratio (risk score[b]) for covariate pattern $\underset{\sim}{x}$.

[a] intro-to-survival-analysis.qmd#def-risk-set
[b] proportional-hazards-models.qmd#def-risk-score

[142] proportional-hazards-models.qmd#def-pha
[143] proportional-hazards-models.qmd#thm-ph-cuhaz
[144] proportional-hazards-models.qmd#thm-loghaz-decomp
[145] Intro-to-GLMs.qmd#def-linear-predictor
[146] proportional-hazards-models.qmd#def-risk-score
[147] intro-to-survival-analysis.qmd#def-hazard-ratio
[148] proportional-hazards-models.qmd#thm-hazard-ratio-ph

Exm

Example 0.1 (Breslow estimator with two event times)**.** Suppose there are two distinct event times $t_1 = 2$ and $t_2 = 4$ (no ties, $d_1 = d_2 = 1$), with risk-set-weighted totals $\sum_{k \in R(2)} \theta_\lambda(\underset{\sim}{x}_k) = 10$ and $\sum_{k \in R(4)} \theta_\lambda(\underset{\sim}{x}_k) = 6$. Then

$$\hat{\Lambda}_0(4) = \frac{1}{10} + \frac{1}{6} \approx 0.267.$$

Akaike, Hirotugu. 1974. "A New Look at the Statistical Model Identification." *IEEE Transactions on Automatic Control* 19 (6): 716–23. https://doi.org/10.1109/TAC.1974.1100705.

Schwarz, Gideon. 1978. "Estimating the Dimension of a Model." *The Annals of Statistics* 6 (2): 461–64. https://doi.org/10.1214/aos/1176344136.